Neutron Streaming Through Gaps in Fusion Reactor Shielding: Revisiting a 1984 Study with Modern Monte Carlo and Weight-Window Variance Reduction

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Abstract

The 1984 study of Halley, later summarized with Miller in a 1986 journal article, used the Monte Carlo code MORSE-CG to characterize 14-MeV neutron streaming through straight and stepped gaps between the outboard shield sectors of the STARFIRE conceptual tokamak. We independently reproduce the principal conclusions of that work with the modern open-source Monte Carlo code OpenMC (v0.15.3) and ENDF/B-VIII.0 cross sections, and extend it by applying mesh-based weight-window variance reduction (the MAGIC method) to converge the deep-penetration quantities — the solid-shield background and the stepped-gap leakage — that were below the analog sampling floor in the original work, reaching sub-2% statistical precision with roughly 50–70 times fewer histories than analog sampling would require. The straight-slot transverse dose profile reproduces the original Fig. IV-1 quantitatively; the three-step configuration reproduces the original Fig. IV-11 result (leakage within an order of magnitude of the unbroken shield); and the single-step configuration reproduces the qualitative two-peak structure of the original Fig. IV-8, with the quantitative differences traced to the original's under-specified step geometry and high analog variance. Mesh-based variance reduction brings the deep-penetration regime — the part of the problem most relevant to shield-design margins — within routine reach of open-source Monte Carlo.

Keywords — neutron streaming, Monte Carlo, weight-window variance reduction, fusion reactor shielding, OpenMC

I. INTRODUCTION

Conceptual tokamak power-plant designs build the plasma chamber and radiation shield in pie-shaped sectors, leaving thermal-expansion gaps between adjacent sectors. Such gaps are streaming paths: 14-MeV fusion neutrons can penetrate the shield far more easily through a gap than through the bulk material, producing locally elevated dose behind the gap and activating components the shield is meant to protect. Offsetting or stepping the gap is the standard mitigation, and quantifying how much a step buys is a basic shielding-design question.

STARFIRE — a 1200-MWe commercial tokamak study by Argonne National Laboratory and partners (1980) — was used as the reference design for the 1984 streaming study because its shield configuration and materials were specified in detail. That study [1, 2] used MORSE-CG, a multigroup Monte Carlo code, with the FLUNG 35-group fusion cross-section library, and a two-step "MORSE-to-MORSE" coupling: the first calculation transported neutrons from the plasma through the blanket, reflector, and plenum to the shield jacket to build a degraded surface ("pseudo-") source; the second transported that source through the shield gap to point detectors outside.

A recurring difficulty in such calculations is that the quantities most relevant to design margins — the dose behind an unbroken shield, and the residual leakage through a well-stepped gap — lie many decades below the source and are therefore hard to resolve by analog Monte Carlo. In the original work the unbroken-shield background was reported with a fractional standard deviation (FSD) above 3, i.e. it was not a converged result. Modern variance-reduction methods, unavailable in their mature mesh-based form in 1984, make these quantities accessible, and revisiting the problem is a natural way to demonstrate that.

We pursue two aims: (i) independently reproduce the original study's principal conclusions with a continuous-energy Monte Carlo code and modern nuclear data; and (ii) apply weight-window variance reduction to converge the deep-penetration quantities — the unbroken-shield background and the stepped-gap leakage — that analog Monte Carlo cannot resolve to design-relevant precision, and quantify the resulting efficiency gain.

Throughout, "reproduction" means independent reproduction of the original study's principal conclusions, not numerical replication: the present work uses a different code, nuclear-data library, steel composition, source treatment, and dose-conversion factors, and transports neutrons only.

II. METHODS

Figure 1 summarizes the workflow relating the original 1984 calculation to the present reproduction: the original two-step MORSE-CG coupling (§2.1) is superseded here by a single-pass OpenMC slab model with a degraded-spectrum surrogate source (§2.2), converged with MAGIC weight-window variance reduction (§2.3) to obtain the production results of §3.

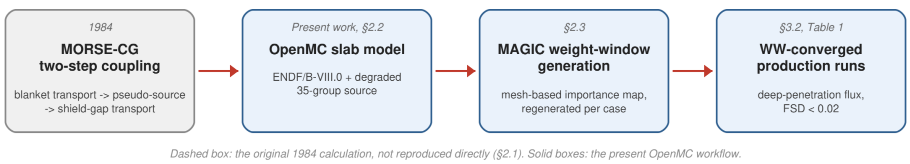

Fig. 1. Workflow schematic. The original 1984 MORSE-CG two-step coupling (dashed, historical, §2.1) is not reproduced directly; the present work instead builds an OpenMC slab model with a degraded-spectrum surrogate source (§2.2) and applies MAGIC weight-window variance reduction (§2.3) to converge the deep-penetration production runs reported in §3.2.

II.A. The original calculation (summary)

The original study [1, 2] used MORSE-CG, a multigroup three-dimensional Monte Carlo neutron and gamma-ray transport code, with the FLUNG library — a 35-neutron-group, 21-gamma-group, P_3 cross-section set (DLC-86) collapsed from the 171-group VITAMIN-C library for fusion neutronics. Flux-to-dose conversion used the ANSI/ANS-6.1.1-1977 [3] factors, fitted to the 35-group structure. The toroidal reactor was idealized to a rectilinear "wedge," 200 cm $\times$ 200 cm in cross-section and 573 cm deep, with albedo (reflecting) boundaries on the transverse faces to represent the repeating sector pattern.

Because a single transport calculation from the plasma to detectors outside the shield is too deep to sample directly, the study used a two-step "MORSE-to-MORSE" coupling. The first step transported a homogeneous, isotropic, multi-energy volumetric source ($\approx$ 80% of its weight in the 14-MeV group; the spectrum of Table III-3) from the plasma through the first wall, neutron multiplier, breeding blanket, reflector, and plenum to a point on the shield jacket, recording each neutron's energy, statistical weight, and direction to build a degraded surface ("pseudo-") source.

The second step launched that pseudo-source through the outboard bulk shield — high-, medium-, and low-flux zones (0.5 / 0.4 / 0.18 m, per the author's 1984 thesis [2], the primary source for the original calculation) — and to point detectors at the shield surface and 50 cm beyond it, with dose reported versus transverse position. The unbroken-shield background was reported as $\approx 1 \times 10^{-33}$ rem·hr^{-1} per source neutron with a fractional standard deviation above 3 — i.e., not a converged value. Straight slots (1, 2, 5, 10 cm) and stepped configurations (single steps of 3–10 cm center-to-center, and a three-step staircase) were examined; the sector joint that these idealize is a multi-step castellated labyrinth (Fig. II-5 of the original), for which "no detailed geometrical design was specified."

II.B. OpenMC model

All calculations use OpenMC 0.15.3 [4] with ENDF/B-VIII.0 cross sections [5], obtained as the official neutron sublibrary release from the National Nuclear Data Center (nndc.bnl.gov), processed with NJOY2016 (B-10 taken from the corresponding official ENDF/B-VII.1 release owing to an NJOY2016 processing incompatibility in the VIII.0 evaluation), run on the Google Colab free tier.

Geometry. The toroidal shield is idealized to a rectilinear slab, 200 cm in the transverse (x) direction and 100 cm in height (y), both with reflective boundaries; the beam axis is z. The original used albedo boundaries on these transverse faces (§2.1); a reflective (albedo = 1) boundary returns more transverse flux than an albedo < 1 would, so this substitution likely overestimates transverse return somewhat. We bound this effect directly: rerunning the two Table 1 deep-penetration cases (solid shield, three-step gap) with vacuum (albedo = 0) transverse boundaries in place of reflective gives 3.62×10^{-10} (FSD 0.020) and 4.61×10^{-10} (FSD 0.022) respectively — 0.17 and 0.14 decade below the reflective Table 1 values, statistically significant at 12.4σ and 9.9σ respectively but small relative to the decade-scale streaming comparisons drawn throughout this study. Since reflective (albedo = 1) and vacuum (albedo = 0) bracket the true boundary condition, these two runs give a rigorous two-sided bound rather than a one-sided sensitivity check. The shield is the three-zone outboard bulk shield: high-flux (HFS, 50 cm), medium-flux (MFS, 40 cm), and low-flux (LFS, 18 cm) zones, 108 cm total. The LFS thickness follows the author's 1984 thesis [2] rather than a divergent 0.28 m figure listed for this region in Table III-1 of the STARFIRE design report [6];

the thesis is the primary source for the neutron-transport model reproduced here, and we adopt it as authoritative. A source void ($z = -20$ to 0) precedes the shield and a detector void ($z = 108$ to 160) follows it, both vacuum-terminated. Gaps are modeled as continuous void channels of half-width w through the shield, in three configurations defined next.

Gap configurations. A *straight slot* runs un-offset along z at $x \in [-w, w]$. A *three-step* gap is offset laterally by a fixed step s at each shield-zone interface, so its HFS, MFS, and LFS segments are centered at $x = 0, +s$, and $+2s$; for the configuration reported here $w = 0.5$ cm and $s = 5$ cm, so adjacent segments do not overlap and no straight-line path traverses the shield. A *single-step* gap runs straight through the inboard half of the shield and then jogs laterally by one step ($\approx$ 3 cm center-to-center) for the outboard half, the two legs joined by a short transverse connector so that the void channel is continuous — a dog-leg, matching the continuous-labyrinth character of the physical sector joint rather than two disconnected slots. Straight slots of 1 and 10 cm full width are reported here (§3.1, §3.2). Figure 2 shows the three configurations in cross-section.

Materials. The three shield zones are homogenized mixtures specified by volume fraction from the original design: HFS = 5% Ti-6Al-4V + 65% TiH_2 + 15% B_4C + 15% H_2O; MFS = 70% Fe-1422 + 15% B_4C + 15% H_2O; LFS = 100% Fe-1422. Fe-1422 is the STARFIRE bulk-shield steel, a low-activation formulation selected in the original design to limit long-lived-isotope production (the outboard shield was required to use materials of low long-lived radioactivity), with its detailed composition given in the STARFIRE report [6]. We approximate it here by a 316-type austenitic stainless steel at 8.0 g cm^{-3}; this is a convenience that over-represents nickel relative to the low-activation original. Bulk-shield attenuation at these energies is governed by iron, which the two compositions share, so we expect this substitution to be a second-order effect on the streaming ratios and variance-reduction results reported here; a macroscopic cross-section comparison against two published low-activation steels, quantifying this at the percent level, is reported in §4. An exact Fe-1422 composition would refine the absolute MFS and LFS attenuation and is the principal residual material uncertainty (§4).

Source. A 35-group degraded spectrum (the original Table III-3, ∼80% weight in the 14-MeV group) is emitted from a plane at the shield front into the forward hemisphere (isotropic in the $+z$ half-space). This is a lightweight surrogate for the original's coupled pseudo-source; a full two-step coupling was not reproduced.

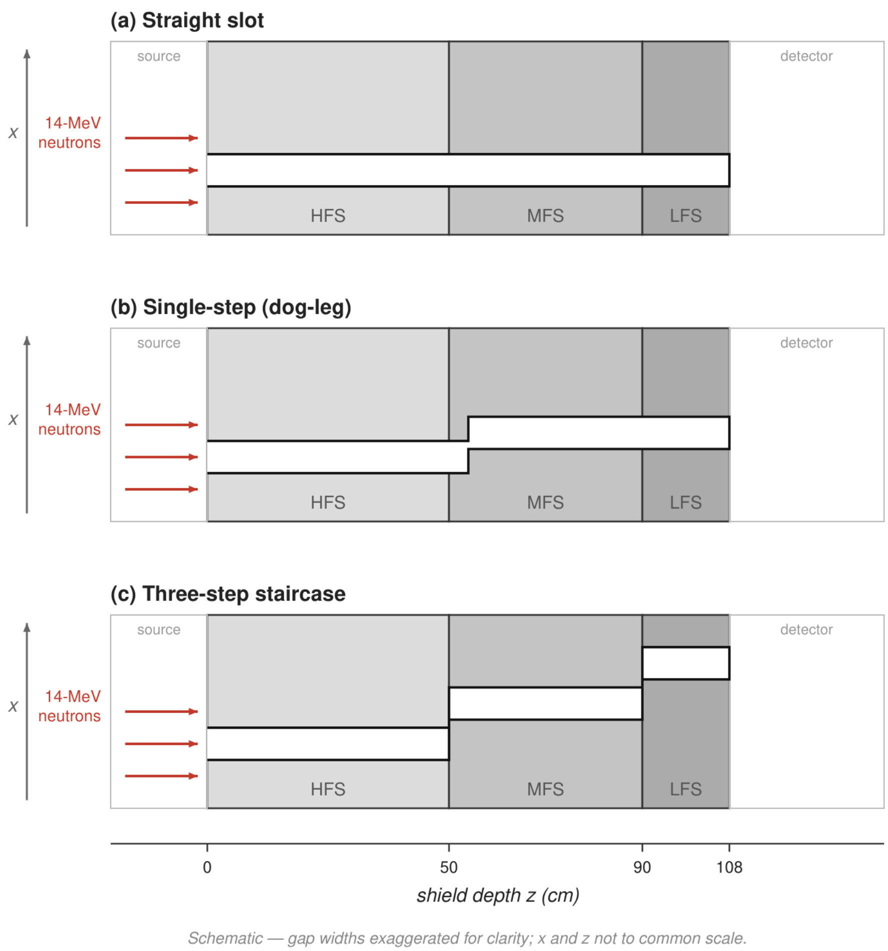

Fig. 2. Cross-sectional schematic of the three gap configurations, in the shield-depth (z) – transverse (x) plane: **(a)** a straight slot running un-offset through all three shield zones (HFS, MFS, LFS); **(b)** a single-step dog-leg, straight through the inboard half and jogged laterally by one step for the outboard half; and **(c)** a three-step staircase, offset by one step at each zone interface so that no straight-line path traverses the shield. 14-MeV neutrons are incident from the left; a source void precedes the shield and a detector void follows it. Gap widths are exaggerated for visibility and the transverse and depth axes are not to a common scale.

Tallies. This reproduction transports neutrons only; secondary photons are not generated or tracked, unlike the original's coupled neutron-gamma MORSE-CG/FLUNG calculation. Two observables are used: (i) a cell tally of total flux in the detector void, volume-averaged over DET_VOLUME = $200 \times 100 \times 52 = 1.04 \times 10^6$ cm^3, for integral comparisons; and (ii) a fine transverse mesh (0.5 cm x-bins) of neutron dose (ICRP-116 [7] anterior-posterior coefficients) at the shield surface and at +50 cm, for the dose-versus-position profiles that the original figures report. Transverse-profile values reported below in "detector units" (§3.1, §3.3–3.4) are this dose-mesh tally's native OpenMC output: the track-length flux in each mesh cell weighted by the ICRP-116 dose coefficient (pSv·cm^2) and reported per source neutron, without volume-averaging the mesh cell or scaling to an absolute source strength. Because every bin in a given figure shares the same mesh-cell volume, ratios and decade-scale falloffs between bins are physically meaningful even though the absolute value is a tally-normalization convention rather than a literal dose rate. The ICRP-116 anterior-posterior (AP) coefficients used here presume a mono-directional field incident on a standard phantom's front; the actual field at these mesh tallies is not mono-directional, so "AP dose" is a labeling convention carried over from the coefficient set's name, not a literal AP-geometry irradiation calculation — the quantity of substance is the dose-coefficient-weighted fluence used consistently for all shape comparisons, not an absolute personal-dose estimate. The original's Fig. IV-1/IV-8/IV-11 dose curves are total (n+γ) dose; behind a meter of steel-dominated shield the secondary-gamma component is not necessarily negligible relative to the neutron component. The comparisons in §3 are therefore neutron-dose-shape comparisons against total-dose curves, which we judge adequate for the decade-scale falloffs and ratios reported here (the gamma component should vary comparatively smoothly with position relative to the neutron streaming signature that is the object of comparison), but this is an assumption, not a demonstrated equivalence; a coupled n-γ rerun would close this gap directly (§4). The original used the ANSI/ANS-6.1.1-1977 [3] flux-to-dose factors, which differ from ICRP-116 in a spectrum-dependent way; because the incident (14-MeV-dominated) and behind-shield (degraded) spectra differ, we do not expect this difference to cancel exactly, but judge it a second-order effect relative to the decade-scale comparisons drawn here.

II.C. Weight-window variance reduction

Deep-penetration quantities — the unbroken-shield background and the stepped-gap leakage — converge poorly under analog sampling: analog runs of 10^7 histories (8.3 and 9.2 minutes wall time respectively) give FSD = 0.142 (solid shield) and FSD = 0.127 (three-step gap), an order of magnitude worse than the FSD < 0.02 weight-window results reported below (Table 1). Survival biasing was off in all analog runs, so every history carries unit weight and no single track can contribute a disproportionate score to the tally; for the solid-shield case, the recorded track length implies roughly 110 detector-entering histories over 10^7 source particles, and a naive equal-weight estimate from $\sim$110 such events would itself give FSD ≈ 0.10 — consistent with the measured 0.142 once ordinary score-spread is included, rather than an estimate resting on a handful of outlier events. For the solid-shield case we additionally validated this analog baseline two ways: an independent-seed replicate (a second analog run with a different RNG seed, same 10^7 histories) agreed with the first within 1.2σ (5.078×10^{-10}, FSD 0.142, vs. 6.417×10^{-10}, FSD 0.131), and a batch-checkpointed trajectory of the first run (recorded at 40, 80, 120, 160, and 200 batches) showed the running mean converging toward the weight-window-converged value with a smooth overshoot-and-correct pattern rather than a discontinuous jump — the signature that would indicate a rare, high-weight history had not yet been sampled. These checks support treating the solid-shield analog FSD as a genuine characterization of the sampling uncertainty. The three-step case received the same two-seed treatment; the resulting bracket on the weight-window value, and what it implies for the history-count multiplier, is reported together with the full comparison in §3.2 rather than repeated here. We generate mesh-based weight windows with OpenMC's MAGIC method [8] (`WeightWindowGenerator`, iterated over batches), then run production calculations with the windows applied. MAGIC was chosen over global deterministic-adjoint methods such as CADIS [9] and FW-CADIS [10] because it is implemented natively in OpenMC and requires no external deterministic transport solve to build the adjoint source — a mesh-based, Monte-Carlo-only workflow consistent with this paper's broader point that deep-penetration variance reduction is now within routine reach of open-source Monte Carlo alone, without a separate deterministic code. An unbiasedness guardrail is enforced first: the straight 1-cm case is run with and without weight windows and required to agree within statistics before any deep-penetration result is trusted (§3.2).

The weight-window mesh spans the full source-to-detector region (a regular mesh, 20 $\times$ 1 $\times$ 90 cells — 10-cm bins in the transverse x-direction, a single bin in y consistent with the slab's infinite-in-y idealization, and 2-cm bins along the beam axis z) with a single energy bin (0–14.1 MeV); the weight windows are therefore spatially resolved but not energy-dependent, which is a simplification relative to a fully energy-and-space-resolved variance-reduction scheme and could under-optimize the windows for the softened, down-scattered part of the spectrum that dominates at depth. Generation-phase runs use 0.8–1.2 $\times 10^6$ histories total (20,000 particles per batch, 40–60 batches with a MAGIC-iteration cap of 20–30 realizations depending on case, more for the deeper solid-shield case) to build the mesh; production runs then use 200 batches $\times$ 50,000 particles = 1 $\times 10^7$ histories under the converged weight windows to obtain the values in Table 1. The weight-window-enabled production run therefore uses no more raw histories than an unconverged analog run of the same size — the efficiency gain is entirely in the per-history variance, not in running more particles.

III. RESULTS

III.A. Straight-slot transverse profile (cf. original Fig. IV-1)

For a 10-cm straight slot, the transverse dose profile at the shield surface is flat across the slot mouth ($|x| < 5$ cm, $\sim 7.1 \times 10^{-2}$ in detector units, uniform to $\pm 5\%$), drops at the geometric edge ($x = 5$ cm), and falls **1.08 decades from the edge to $x = 10$ cm** — matching the original's "about one order of magnitude at $x = 10$." All bins have relative error ≤ 0.16. The 1-cm slot, being more collimated, falls off too steeply for analog statistics to trace into the wings (it is resolved only to $x \approx 1$ cm), but its surface-versus-+50 cm behavior reproduces the original's beam-spread observation: on-axis dose is nearly flat with distance while off-axis dose rises, the signature of a diverging pencil beam.

III.B. Deep-penetration quantities via weight-window variance reduction

The unbiasedness guardrail passes: the straight 1-cm case gives 2.50 $\times 10^{-9}$ with weight windows (FSD 0.052) versus a prior validated analog run of the same model (10^7 histories, 2.32 $\times 10^{-9}$, FSD 0.069; ratio 1.078, within the pass criterion of 3 $\times$ $\mathrm{FSD}_{WW} = 0.16$). This guardrail exercises the case least likely to reveal weight-window bias, since the straight slot is dominated

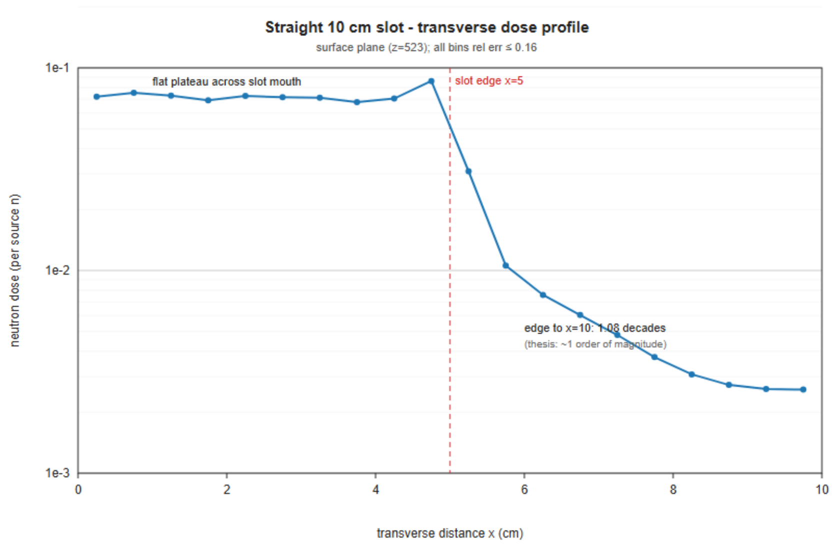

Fig. 3. Transverse dose profile, 10-cm straight slot at the shield surface. Flux is flat across the slot mouth ($|x| < 5$ cm) and falls approximately one decade from the geometric edge to $x = 10$ cm, reproducing the original Fig. IV-1.

by direct streaming that analog sampling already resolves well. For the deep-penetration cases an analog reference is available but imprecise (§2.3: FSD $\approx$ 0.13–0.14 at 10^7 histories). For the solid shield, the weight-window value agrees with both independent analog seeds within their combined statistics (5.37×10^{-10} vs. 5.078×10^{-10}, FSD 0.142, and vs. 6.417×10^{-10}, FSD 0.131 — 0.4σ and 1.2σ respectively). For the three-step gap, the weight-window value (6.42×10^{-10}) and the first analog run (4.678×10^{-10}, FSD 0.127) disagree at 2.9σ. A second, independent-seed analog run gave 7.312×10^{-10} (FSD 0.127), agreeing with the weight-window value within 0.95σ; the two analog seeds disagree with each other at 2.4σ — more than either run's own quoted FSD would predict for two draws from the same distribution — but their average (5.995×10^{-10}) agrees with the weight-window value within 0.8σ. This pattern is consistent with the single-run analog FSD understating the true sampling uncertainty for this quantity, a known risk for a deep-penetration tally fed by rare, multiply-scattered paths, rather than with any bias in the weight-window result: the two independent draws bracket the weight-window value from opposite sides, and averaging resolves most of the apparent tension. We do not attempt to assign a corrected single-run FSD from only two draws, but note that if the true per-run uncertainty exceeds 0.127, the resulting history-count multiplier ($\sim$50$\times$) is, if anything, an underestimate of the true speed-up for this case, consistent with the direction of the solid-shield result rather than in tension with it. We additionally rely on each run's internal batch-to-batch convergence and the built-in MAGIC iteration schedule (20–30 realizations depending on case, chosen so the weight-window mesh itself has stabilized before the production run). As a separate check of the weight-window result's own robustness (independent of the analog-baseline comparison above), both deep-penetration cases were rerun end-to-end (independent weight-window generation and production) with a different RNG seed than the one used elsewhere in this study. This weight-window replicate agreed with the original weight-window run to within 0.09σ for both the solid shield (5.355×10^{-10}, FSD 0.026, vs. 5.370×10^{-10}, FSD 0.017) and the three-step gap (6.439×10^{-10}, FSD 0.026, vs. 6.420×10^{-10}, FSD 0.018) — σ computed as the quadrature combination of both runs' absolute standard deviations — confirming the weight-window-converged values are not an artifact of the particular RNG stream used. Weight windows then converge the two quantities to a precision unmeasurable either in the 1984 study or in our own single-run analog checks:

TABLE I
Weight-window-converged deep-penetration flux values (all values track-length estimates, per source neutron).

Case	Volume-averaged flux (n·cm^{-2} per source n)	FSD	Wall time
Straight 1 cm	2.50×10^{-9}	0.052	66 min
Unbroken shield (solid)	5.37×10^{-10}	0.017	60 min
Three-step gap (1 cm, 5-cm offset)	6.42×10^{-10}	0.018	62 min

The present work provides a converged Monte Carlo value for the unbroken-shield penetration flux of this STARFIRE configuration; analog Monte Carlo resolves it only to FSD = 0.142 at 10^7 histories (§2.3), a result validated by an independent-seed replicate and a batch-checkpointed convergence trajectory showing no discontinuous jump. The weight-window solid-shield run reaches FSD = 0.017 at the wall time in the table above. Since FSD scales as $N^{-1/2}$, reaching FSD = 0.017 by analog sampling alone, starting from the measured analog FSD = 0.142 at 10^7 histories, requires roughly $(0.142/0.017)^2 \approx 70$ times more histories for the solid-shield case; the analogous calculation for the three-step case (measured analog FSD = 0.127 at 10^7 histories) gives roughly $(0.127/0.018)^2 \approx 50$ times more histories. We report this history-count ratio as the primary efficiency figure because it follows directly from Monte Carlo statistics (FSD $\propto N^{-1/2}$) independent of hardware or implementation. The wall-clock ratio we actually measured is more conservative: weight-window production runs took roughly 6–8× longer per history than analog on this hardware, owing to splitting/Russian-roulette bookkeeping overhead, giving a true figure-of-merit gain of roughly 7–10× on the Colab free tier specifically — a practical figure that would be expected to vary with different hardware or a different weight-window implementation, unlike the history-count ratio above. Figure 4 shows the three converged values; the unbroken-shield and three-step quantities, plotted here with sub-2% statistical error, are the two for which analog sampling alone remains comparatively imprecise at equal history count.

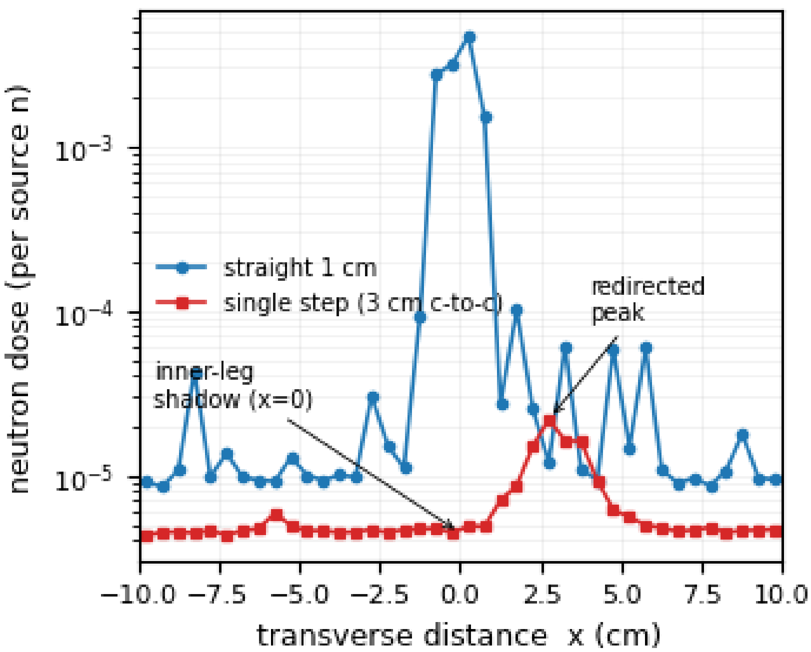

Fig. 4. Three converged deep-penetration flux values with weight-window variance reduction. The unbroken-shield and three-step quantities (FSD < 0.02) are an order of magnitude more precise than the corresponding analog values (FSD $\approx$ 0.13–0.14 at equal history count).

III.C. Three-step gap (cf. original Fig. IV-11)

The three-step staircase transmits 6.42×10^{-10} versus the solid 5.37×10^{-10} — a ratio of 1.19, i.e. **within 0.08 decade of the unbroken shield**. This reproduces the original Fig. IV-11 finding that a three-step gap "reduces the dose rates to within an order of magnitude of the dose of the uninterrupted shield." Physically, a multi-step gap with offset larger than its width has no through line-of-sight; the streaming pencil beam is removed and the detector sees essentially the diffuse shield background. The integral leakage is not reduced five decades because a step de-collimates rather than absorbs — total leakage is approximately conserved and redistributed, a point made quantitative in §3.4.

The transverse dose profile (Fig. 5) makes this de-collimation visible directly: rather than the sharp, narrow peak at $x = 0$ seen for a straight slot (§3.1), the three-step case shows a broad, redistributed profile peaking off-axis at $x = 5.0$ cm — the transverse position of the offset MFS gap segment (the LFS segment itself sits at $x = 10$ cm) — with peak dose 1.483 $\times 10^{-4}$ at the shield surface and 8.989 $\times 10^{-5}$ at +50 cm (both per source neutron, in detector units). The peak sits over the MFS segment rather than the LFS exit aperture because a neutron reaching the LFS void must first cross the HFS→MFS corner and then the MFS→LFS corner; direct penetration through the 18-cm LFS steel from the MFS void is evidently a shorter path than negotiating the second corner, so the MFS gap position dominates the exit-plane dose. The absence of a sharp on-axis peak, together with the modest surface-to-+50 cm falloff (a factor of 1.65), is consistent with a diffuse, non-collimated leakage field rather than a residual direct-streaming path.

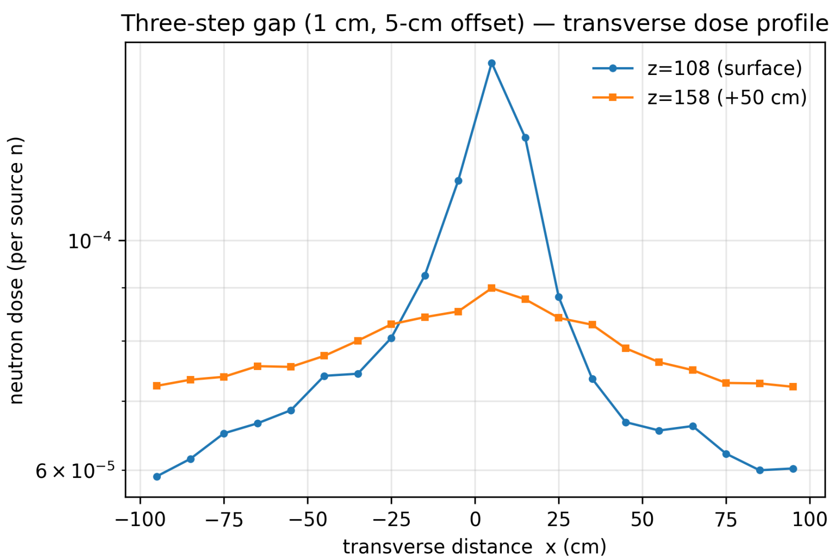

Fig. 5. Transverse dose profile, three-step gap (1 cm, 5-cm offset), at the shield surface ($z = 108$ cm) and +50 cm beyond it ($z = 158$ cm). Unlike the straight-slot profile (Fig. 3), the peak is broad and off-axis ($x = 5.0$ cm, the offset MFS segment) rather than a sharp on-axis streaming peak, reflecting the de-collimation discussed above.

III.D. Single-step gap (cf. original Fig. IV-8)

A single ∼3-cm (center-to-center) step reproduces the **qualitative two-peak structure** of Fig. IV-8: at the +50 cm plane the inner-leg centerline ($x = 0$) is a deep shadow and a redirected peak appears at the outer-leg position ($x \approx 2.8$ cm). Quantitatively, the inner-leg shadow is 2.85 decades below the straight-slot peak (versus the original ∼5 decades) and the redirected peak is 214× below it (versus the original ∼4×). All values are well-converged ($x = 0$ relative error 0.04).

We attribute the quantitative difference to the original's under-specified step geometry and its high analog variance (FSD 0.7–1.3), not to any difference in method (both this work and the original are Monte Carlo); §4 takes this up in detail. The single-step comparison is thus a qualitative reproduction with a precise modern value, rather than a quantitative match to an under-specified, high-variance original.

Integrated over the exit plane, however, the stepped configurations leak within a factor of a few of a straight slot of the same width (§3.2: 6.42×10^{-10} for the three-step gap versus 2.50×10^{-9} for the straight 1-cm slot). A step de-collimates rather than absorbs: total leakage is approximately conserved and redistributed in angle and position, so the point dose on the original beam axis collapses by decades while the integral barely moves. The deep shadow and the redirected peak of Fig. 6 are the two faces of this redistribution.

IV. DISCUSSION

What reproduces, and at what level. Table II summarizes the comparisons at a glance.

Two of the original's results reproduce quantitatively. The straight-slot transverse falloff (§3.1) matches Fig. IV-1 — a flat-topped profile across the slot mouth and roughly one decade of falloff from the slot edge to $x = 10$ cm. The three-step gap (§3.3) reproduces Fig. IV-11: leakage within an order of magnitude of the unbroken shield (here, within 0.08 decade). The single-step result (§3.4) reproduces the *qualitative* structure of Fig. IV-8 — a deep shadow at the inner-leg centerline and a redirected peak at the offset leg — but not the original's quantitative five-decade shadow and four-fold redirected peak.

Why integral and point metrics tell different stories. As §3.4 makes quantitative, a step de-collimates the streaming beam rather than absorbing it, so the volume-integrated stepped-gap flux stays close to the straight-gap flux while the *point* dose on the original beam axis falls

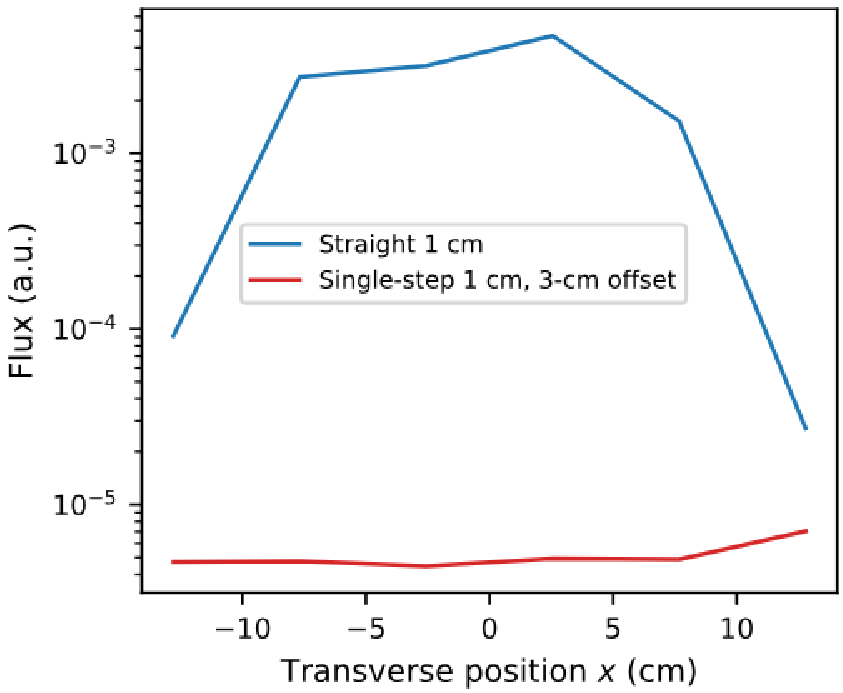

Fig. 6. Transverse flux profile at the +50 cm plane, straight 1-cm slot versus single-step 1-cm slot with 3-cm offset. The shadow at $x = 0$ and the redirected peak at $x \approx 2.8$ cm reproduce the qualitative two-peak structure of the original Fig. IV-8.

TABLE II
Summary comparison, 1984 MORSE-CG original vs. present OpenMC reproduction.

Quantity	1984 MORSE-CG	Present (OpenMC)	Agreement
Straight-slot transverse falloff (Fig. IV-1)	∼1 decade, edge to $x =$ 10 cm	1.08 decades	Quantitative match
Unbroken-shield background flux	$\approx 1 \times 10^{-33}$ rem·hr^{-1} per source n (FSD > 3, not converged)	5.37 $\times 10^{-10}$ n·cm^{-2} per source n (FSD 0.017)	Converged Monte Carlo value; units differ (rem·hr^{-1} vs. flux) so this is a convergence comparison, not a magnitude match
Three-step gap leakage vs. solid (Fig. IV-11)	"Within an order of magnitude" of solid shield	0.08 decade from solid shield (ratio 1.19)	Quantitative match
Single-step shadow depth (Fig. IV-8)	∼5 decades below straight-slot peak (FSD 0.7–1.3)	2.85 decades below straight-slot peak	Qualitative match; quantitative offset discussed below
Single-step redirected-peak intensity (Fig. IV-8)	∼4× below straight-slot peak (FSD 0.7–1.3)	214× below straight-slot peak	Qualitative match; quantitative offset discussed below

by many decades. A volume-averaged detector cannot register a collimation change and therefore cannot show the original's five-decade reduction, which is intrinsically a point-detector quantity. This is a feature of the physics, not a disagreement: the integral and point comparisons are answering different questions, and both are reproduced at the level appropriate to each.

The single-step quantitative offset. The residual difference in the single-step values is not a difference in method — both the original and this work are Monte Carlo (the original used MORSE-CG). Two factors account for it. First, the original modeled an idealized single step, whereas the physical sector joint is a multi-step castellated labyrinth for which the corner geometry — which controls how efficiently wall scattering feeds the redirected beam — was never specified. Our square dog-leg connector is one reasonable idealization among many, and the redirected-beam intensity is sensitive to it. Second, the original single-step calculations carried large statistical uncertainty (FSD 0.7–1.3), which the present weight-window calculation removes. The single-step comparison is therefore best read as a precise modern value for a specific idealized geometry, set against an imprecise original for an under-specified one.

Variance reduction as the methodological contribution. Neither the original calculation nor our own analog runs converge the unbroken-shield background or the stepped-gap leakage to high precision; analog sampling resolves them only to FSD ≈ 0.13–0.14 at 10^7 histories (§2.3). Mesh-based weight windows converge them to FSD < 0.02, requiring roughly 50–70 times fewer histories than analog sampling would need to reach the same precision (§2.3, §3.2); the true wall-clock efficiency gain we measured, which also reflects the extra per-history cost of weight-window bookkeeping, is a more modest $\sim$7–10$\times$ on the hardware used here (§3.2). The practical contribution of this reproduction is therefore twofold: a modern open-source code recovers the original results, and routine variance reduction makes the deep-penetration regime — the part of the problem most relevant to shield-design margins — an order of magnitude more precise, at roughly 6–8$\times$ the wall-clock cost, for a net figure-of-merit gain of $\sim$7–10$\times$. This is a statement about method and tooling now in common use, not about the original calculation, which was sound for what analog Monte Carlo could deliver at the time.

Residual uncertainties and limitations. Six caveats bound the present results. (i) The STARFIRE bulk-shield steel Fe-1422 — a low-activation, low-nickel formulation — is approximated by a 316-type stainless steel, which over-represents nickel; the exact composition [6] would refine the MFS/LFS attenuation. We bound this substitution's effect by comparing the macroscopic total cross section of 316-type stainless steel against two published low-activation steels of similar iron content — EUROFER97 [11] and 9Cr-1Mo (T91) [12] — using the same ENDF/B-VIII.0 data as the rest of this study. Both proxies agree with 316-type stainless steel to within 3.5% at 14.1 MeV and 1.6% at 1 MeV, small relative to the decade-scale streaming ratios reported here, supporting treatment of the substitution as a second-order effect. Fe-1422's own composition [6] is not independently available to us, so this is a bound rather than an exact match, and the EUROFER97 comparison omits tungsten, vanadium, and tantalum, and the 9Cr-1Mo model omits T91's vanadium and niobium microalloying (all absent from our cross-section library), a minor further approximation. This uncertainty is inherited by the converged solid-shield background value in Table 1, the paper's flagship deep-penetration result, not only by the streaming ratios. Because attenuation is exponential in the shield's optical depth, a percent-level cross-section offset can compound to a few tenths of a decade in the absolute transmitted flux; the streaming ratios, in which the shield path is common to numerator and denominator, are insensitive to this at leading order. (ii) In place of the original's two-step coupling we apply the degraded Table III-3 spectrum directly at the shield front as a forward-hemisphere surrogate source; this reproduces the incident spectrum but not the exact angular distribution the blanket would impose, and the source's angular distribution — not only corner geometry — plausibly contributes to the single-step quantitative offset discussed above. (iii) The single-step corner geometry is under-specified, as discussed. (iv) The model is a 2-D slab idealization of a 3-D toroidal geometry, as in the original. (v) Only neutrons are transported; the original reported coupled neutron-gamma dose, and we compare neutron-dose shapes to total-dose curves rather than reproducing the gamma component directly (§2.2). (vi) We bounded the weight-window mesh resolution (baseline $20 \times 1 \times 90$ cells, 10-cm transverse and 2-cm axial bins) and MAGIC iteration count (baseline 20–30 realizations) by rerunning both Table 1 deep-penetration cases at a coarser setting ($10 \times 1 \times 45$ cells, 13–15 realizations) and a finer setting ($40 \times 1 \times 180$ cells, 38–45 realizations). All four alternate values agree with the baseline to within 0.7σ (solid shield: -0.007 decade at 0.7σ coarse, -0.005 decade at 0.5σ fine; three-step gap: -0.001 decade at 0.1σ for both), with no significant shift in either direction. This directly confirms the baseline mesh and iteration count are adequate rather than resting on the indirect argument (FSD $\propto N^{-1/2}$ scaling and RNG-seed agreement) used in an earlier draft of this caveat. None of these affects the central variance-reduction result, which is a statement about method, not materials.

V. CONCLUSIONS

Modern continuous-energy Monte Carlo with weight-window variance reduction reproduces the principal 1984 STARFIRE gap-streaming conclusions and converges the deep-penetration quantities — the unbroken-shield background and the stepped-gap leakage — an order of magnitude more precisely than analog sampling achieves at equal computational history count. The straight-slot falloff and the three-step $\approx$ solid result are quantitative reproductions; the single-step five-decade point reduction is reproduced qualitatively, with the quantitative offset traced to the original's idealized, under-specified step geometry and its high analog variance rather than to any difference in method. The principal methodological point is that mesh-based variance reduction now brings deep-penetration shielding quantities within routine reach of open-source Monte Carlo — a capability not available when the problem was first studied.

AUTHOR CONTRIBUTIONS

A. M. Halley is the sole author of the present work: he performed the original 1984 calculations, designed and ran the OpenMC reproduction, produced the figures, and wrote the manuscript.

CRediT statement: **A. M. Halley:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing.

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During the preparation of this work the author used Claude (Anthropic; models claude-sonnet-4-6, claude-opus-4-8, claude-sonnet-5, and claude-fable-5) to assist with code development, debugging of the OpenMC model and weight-window workflow, and manuscript editing, in order to accelerate an independent one-person reproduction of a 40-year-old calculation for which no original code survives. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article. No AI-generated content was used in the creation of figures, which were produced by Python scripts run by the author.

DECLARATION OF INTEREST

The author declares no competing interests.

DATA AVAILABILITY

The complete OpenMC model and weight-window generation and production cells are in the public repository `github.com/alanhalle/openmc-starfire`, runnable end-to-end on the Google Colab free tier; the statepoints are regenerated by running the notebook. A citable archived release (v1.1.0, matching this submission) is available at https://doi.org/10.5281/zenodo.21418400 (all versions: https://doi.org/10.5281/zenodo.20788782).

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